

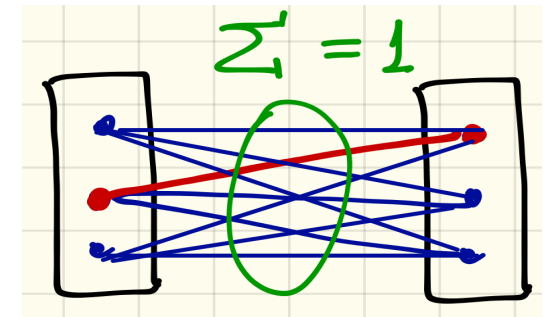
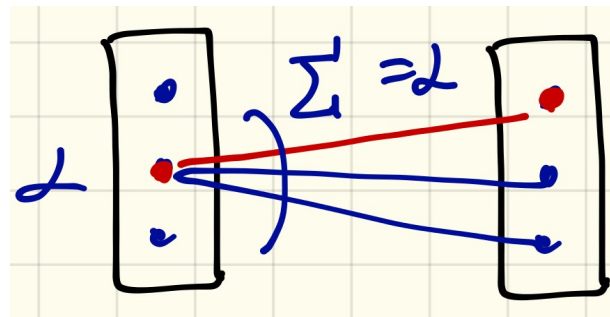
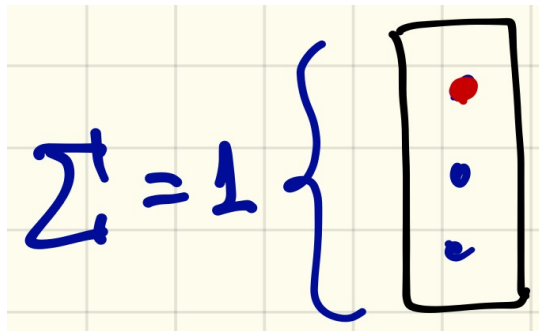
Recall: Local Polytope Relaxation

$$\min_{\mu} \langle \theta, \mu \rangle$$

$$1 - \sum_{x_v \in \mathcal{X}_v} \mu_v(x_v) = 0, \quad v \in \mathcal{V}$$

$$1 - \sum_{(x_u, x_v) \in \mathcal{X}_{uv}} \mu_{uv}(x_u, x_v) = 0, \quad uv \in \mathcal{E}$$

$$\text{s.t. } \sum_{x_v \in \mathcal{X}_v} \mu_{uv}(x_u, x_v) - \mu_u(x_u) = 0, \quad x_u \in \mathcal{X}_u, \quad uv \in \mathcal{E}$$
$$\sum_{x_u \in \mathcal{X}_u} \mu_{uv}(x_u, x_v) - \mu_v(x_v) = 0, \quad x_v \in \mathcal{X}_v, \quad uv \in \mathcal{E}$$
$$-\mu \leq 0.$$



Dual to the Local Polytope Relaxation

$$\min_{\mu} \langle \theta, \mu \rangle$$

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$$\text{s.t.} \quad \sum_{x_v \in \mathcal{X}_v} \mu_{uv}(x_u, x_v) - \mu_u(x_u) = 0, \quad x_u \in \mathcal{X}_u, \quad uv \in \mathcal{E}$$

$$\sum_{x_u \in \mathcal{X}_u} \mu_{uv}(x_u, x_v) - \mu_v(x_v) = 0, \quad x_v \in \mathcal{X}_v, \quad uv \in \mathcal{E}$$

$$-\mu \leq 0.$$

γ_v

γ_{uv}

$\phi_{v,u}(x_v)$

$\phi_{u,v}(x_u)$

$\beta \in \mathbb{R}^I$

Dual to the Local Polytope Relaxation

$$\begin{aligned}
 & \min_{\mu} \langle \theta, \mu \rangle \\
 & \text{s.t.} \quad \begin{aligned}
 & 1 - \sum_{x_v \in \mathcal{X}_v} \mu_v(x_v) = 0, \quad v \in \mathcal{V} \\
 & 1 - \sum_{(x_u, x_v) \in \mathcal{X}_{uv}} \mu_{uv}(x_u, x_v) = 0, \quad uv \in \mathcal{E} \\
 & \sum_{x_v \in \mathcal{X}_v} \mu_{uv}(x_u, x_v) - \mu_u(x_u) = 0, \quad x_u \in \mathcal{X}_u, \quad uv \in \mathcal{E} \\
 & \sum_{x_u \in \mathcal{X}_u} \mu_{uv}(x_u, x_v) - \mu_v(x_v) = 0, \quad x_v \in \mathcal{X}_v, \quad uv \in \mathcal{E} \\
 & -\mu \leq 0.
 \end{aligned}
 \end{aligned}
 \left| \begin{array}{l}
 \gamma_v \\
 \gamma_{uv} \\
 \phi_{v,u}(x_v) \\
 \phi_{u,v}(x_u) \\
 \beta \in \mathbb{R}^I
 \end{array} \right.$$

Lagrangian:

$$\begin{aligned}
 L(\mu, \gamma, \phi, \beta) = & \langle \theta, \mu \rangle - \langle \beta, \mu \rangle + \sum_{v \in \mathcal{V}} \gamma_v \left[1 - \sum_{x_v \in \mathcal{X}_v} \mu_v(x_v) \right] \\
 & + \sum_{uv \in \mathcal{E}} \gamma_{uv} \left[1 - \sum_{(x_u, x_v) \in \mathcal{X}_{uv}} \mu_{uv}(x_u, x_v) \right] \\
 & + \sum_{v \in \mathcal{V}} \sum_{u \in \mathcal{N}_b(v)} \sum_{x_v \in \mathcal{X}_v} \phi_{v,u}(x_v) \left[\sum_{x_u \in \mathcal{X}_u} \mu_{uv}(x_u, x_v) - \mu_v(x_v) \right]
 \end{aligned}$$

Dual to the Local Polytope Relaxation

$$\begin{aligned} g(\mu, \gamma, \phi, \beta) = \inf_{\mu} L(\mu, \gamma, \phi, \beta) &= \sum_{v \in \mathcal{V}} \gamma_v + \sum_{uv \in \mathcal{E}} \gamma_{uv} \\ &+ \sum_{\substack{uv \in \mathcal{E}_a \\ (x_u, x_v) \in \mathcal{X}_{uv}}} \mu_{uv}(x_u, x_v) [\theta_{uv}(x_u, x_v) + \phi_{v,u}(x_v) + \phi_{u,v}(x_u) - \beta_{uv}(x_u, x_v) - \gamma_{uv}] \\ &+ \sum_{v \in \mathcal{V}} \sum_{x_v \in \mathcal{X}_v} \mu_v(x_v) \left[\theta_v(x_v) - \sum_{u \in \mathcal{N}_b(v)} \phi_{v,u}(x_v) - \beta_v(x_v) - \gamma_v \right] \end{aligned}$$

Dual to the Local Polytope Relaxation

$$\begin{aligned}
 g(\mu, \gamma, \phi, \beta) = \inf_{\mu} L(\mu, \gamma, \phi, \beta) &= \sum_{v \in \mathcal{V}} \gamma_v + \sum_{uv \in \mathcal{E}} \gamma_{uv} \\
 &+ \sum_{\substack{uv \in \mathcal{E} \\ (x_u, x_v) \in \mathcal{X}_{uv}}} \mu_{uv}(x_u, x_v) [\theta_{uv}(x_u, x_v) + \phi_{v,u}(x_v) + \phi_{u,v}(x_u) - \beta_{uv}(x_u, x_v) - \gamma_{uv}] \\
 &+ \sum_{v \in \mathcal{V}} \sum_{x_v \in \mathcal{X}_v} \mu_v(x_v) \left[\theta_v(x_v) - \sum_{u \in \mathcal{N}_b(v)} \phi_{v,u}(x_v) - \beta_v(x_v) - \gamma_v \right]
 \end{aligned}$$

Dual problem:

$$\begin{aligned}
 \max_{\gamma, \phi, \beta} \quad & \sum_{v \in \mathcal{V}} \gamma_v + \sum_{uv \in \mathcal{E}} \gamma_{uv} \\
 \gamma_v = & \theta_v(x_v) - \sum_{u \in \mathcal{N}_b(v)} \phi_{v,u}(x_v) - \beta_v(x_v), \quad v \in \mathcal{V}, \quad x_v \in \mathcal{X}_v \\
 \gamma_{uv} = & \theta_{uv}(x_u, x_v) + \phi_{v,u}(x_v) + \phi_{u,v}(x_u) - \beta_{uv}(x_u, x_v), \quad uv \in \mathcal{E}, \quad (x_u, x_v) \in \mathcal{X}_{uv} \\
 \beta \geq & 0
 \end{aligned}$$

Dual to the Local Polytope Relaxation

$$\max_{\gamma, \phi, \beta} \sum_{v \in \mathcal{V}} \gamma_v + \sum_{uv \in \mathcal{E}} \gamma_{uv}$$

$$\gamma_v = \theta_v(x_v) - \sum_{u \in \mathcal{N}_b(v)} \phi_{v,u}(x_v) - \beta_v(x_v), \quad v \in \mathcal{V}, \quad x_v \in \mathcal{X}_v$$

$$\gamma_{uv} = \theta_{uv}(x_u, x_v) + \phi_{v,u}(x_v) + \phi_{u,v}(x_u) - \beta_{uv}(x_u, x_v), \quad uv \in \mathcal{E}, \quad (x_u, x_v) \in \mathcal{X}_{uv}$$
$$\beta \geq 0$$

$$\max_{\gamma, \phi, \beta} \sum_{v \in \mathcal{V}} \gamma_v + \sum_{uv \in \mathcal{E}} \gamma_{uv}$$

$$\gamma_v = \theta_v^\phi(x_v) - \beta_v(x_v), \quad v \in \mathcal{V}, \quad x_v \in \mathcal{X}_v$$

$$\gamma_{uv} = \theta_{uv}^\phi(x_u, x_v) - \beta_{uv}(x_u, x_v), \quad uv \in \mathcal{E}, \quad (x_u, x_v) \in \mathcal{X}_{uv}$$
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Dual to the Local Polytope Relaxation

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$$\beta \geq 0$$

$$\max_{\gamma, \phi, \beta} \sum_{v \in \mathcal{V}} \gamma_v + \sum_{uv \in \mathcal{E}} \gamma_{uv}$$

$$\gamma_v \leq \theta_v^\phi(x_v), \quad v \in \mathcal{V}, \quad x_v \in \mathcal{X}_v$$

$$\gamma_{uv} \leq \theta_{uv}^\phi(x_u, x_v), \quad uv \in \mathcal{E}, \quad (x_u, x_v) \in \mathcal{X}_{uv}$$

Dual to the Local Polytope Relaxation

Constrained dual representation:

$$\max_{\gamma, \phi, \beta} \sum_{v \in \mathcal{V}} \gamma_v + \sum_{uv \in \mathcal{E}} \gamma_{uv}$$

$$\gamma_v \leq \theta_v^\phi(x_v), \quad v \in \mathcal{V}, \quad x_v \in \mathcal{X}_v$$

$$\gamma_{uv} \leq \theta_{uv}^\phi(x_u, x_v), \quad uv \in \mathcal{E}, \quad (x_u, x_v) \in \mathcal{X}_{uv}$$

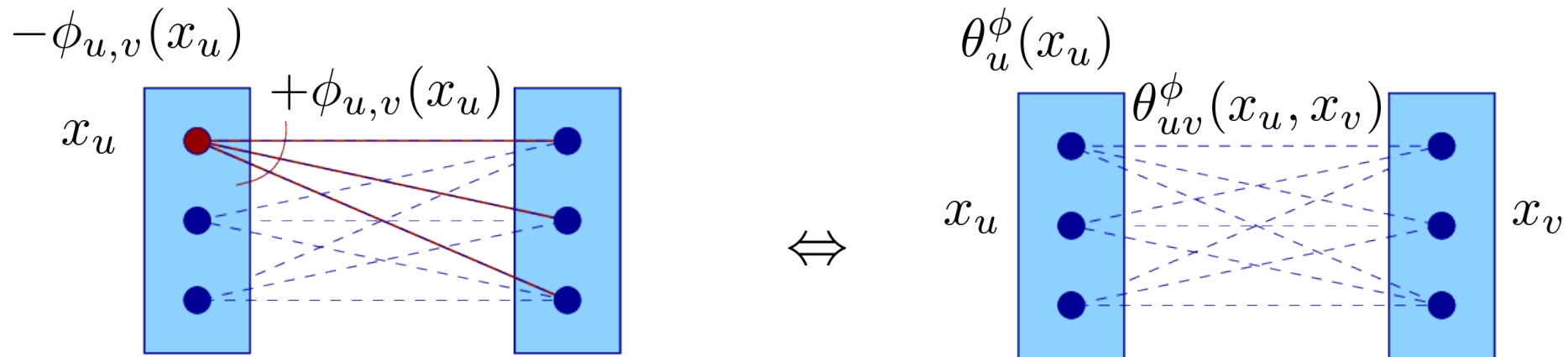
Unconstrained dual representation:

$$\max_{\phi \in \mathbb{R}^J} D(\phi) := \max_{\phi \in \mathbb{R}^J} \sum_{v \in \mathcal{V}} \min_{s \in \mathcal{X}_v} \theta_v^\phi(s) + \sum_{uv \in \mathcal{E}} \min_{(s,t) \in \mathcal{X}_{uv}} \theta_{uv}^\phi(s, t)$$

Concave **piece-wise-linear** (finite number of lines) optimization problem

Reparametrization (Equivalent Transformations)

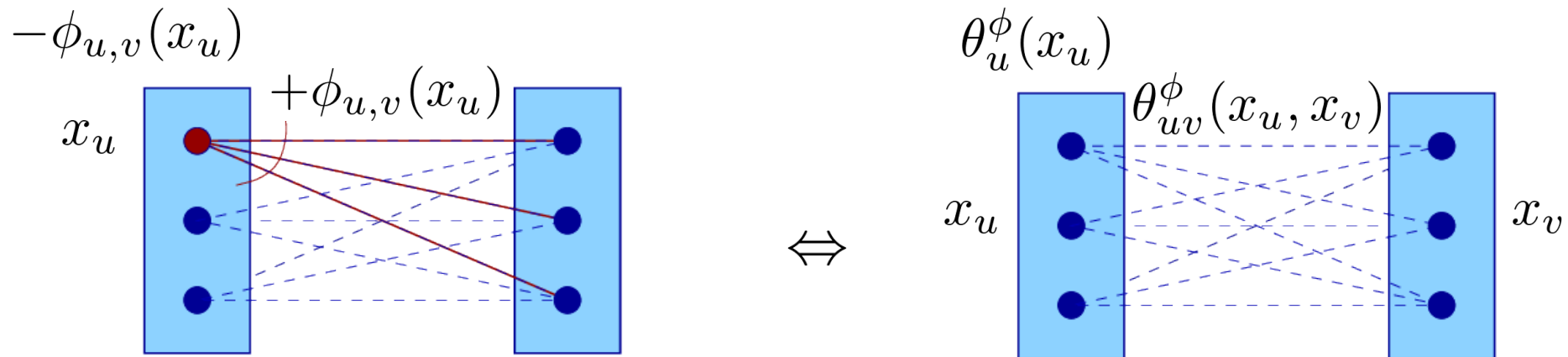
$$\sum_{v \in \mathcal{V}} \theta_v(x_v) + \sum_{uv \in \mathcal{E}} \theta_{uv}(x_u, x_v) = \sum_{v \in \mathcal{V}} \theta_v^\phi(x_v) + \sum_{uv \in \mathcal{E}} \theta_{uv}^\phi(x_u, x_v)$$



Reparametrization (Equivalent Transformations)

$$\sum_{v \in \mathcal{V}} \theta_v(x_v) + \sum_{uv \in \mathcal{E}} \theta_{uv}(x_u, x_v) = \sum_{v \in \mathcal{V}} \theta_v^\phi(x_v) + \sum_{uv \in \mathcal{E}} \theta_{uv}^\phi(x_u, x_v)$$

Proposition: $\langle \theta, \mu \rangle = \langle \theta^\phi, \mu \rangle$ for any $\mu \in \mathcal{L}$



Reparametrization (Equivalent Transformations)

Proposition: $\langle \theta, \mu \rangle = \langle \theta^\phi, \mu \rangle$ for any $\mu \in \mathcal{L}$

Does $\langle \theta, \mu \rangle = \langle \theta^\phi, \mu \rangle$ for any $\mu \in \mathcal{M}$?

- 1) Yes, because $\mathcal{L} \subseteq \mathcal{M}$
- 2) Yes, because $\mathcal{M} \subseteq \mathcal{L}$
- 3) No, because $\mathcal{L} \subseteq \mathcal{M}$
- 4) No, because $\mathcal{M} \subseteq \mathcal{L}$
- 5) Yes, but only for $\mu = \delta(x), x \in \mathcal{X}_{\mathcal{V}}$
- 6) No correct answer

Reparametrization (Equivalent Transformations)

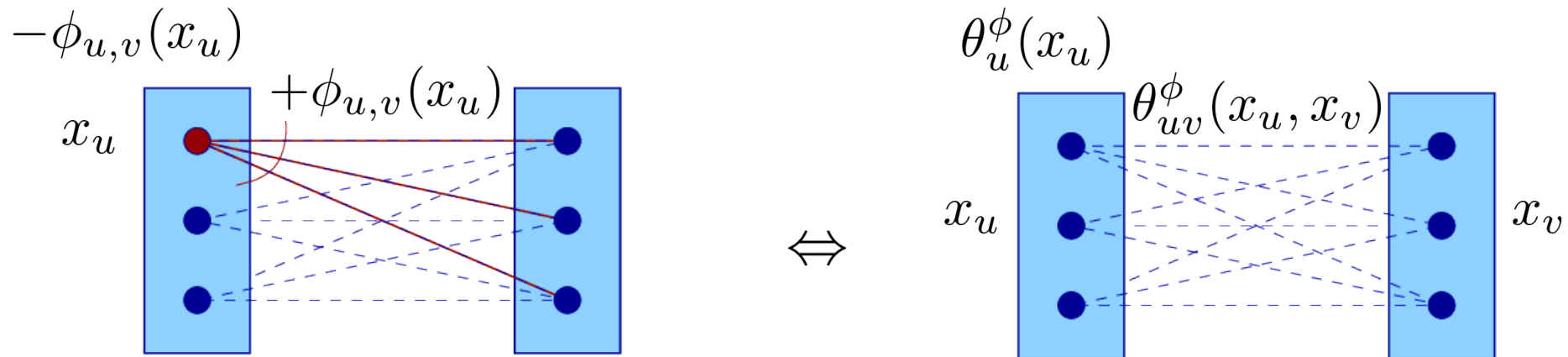
Primal:

$$\min_{x \in \mathcal{X}_{\mathcal{V}}} \left[E(x) := \sum_{v \in \mathcal{V}} \theta_v^\phi(x_v) + \sum_{uv \in \mathcal{E}} \theta_{uv}^\phi(x_u, x_v) \right]$$

Dual:

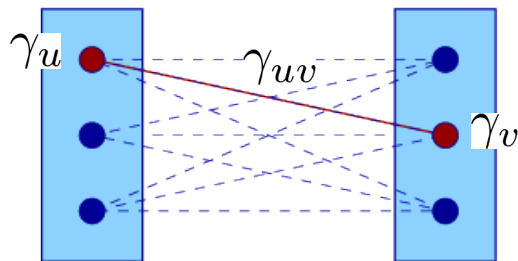
$$\max_{\phi} \left[D(\phi) := \sum_{v \in \mathcal{V}} \min_{s \in \mathcal{X}_v} \theta_v^\phi(x_v) + \sum_{uv \in \mathcal{E}} \min_{(s,t) \in \mathcal{X}_{uv}} \theta_{uv}^\phi(x_u, x_v) \right]$$

$$E(x) \geq D(\phi), \quad \forall x, \phi$$

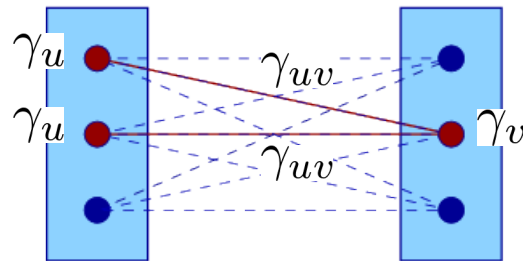


How to Reconstruct a Labeling from a Reparametrization?

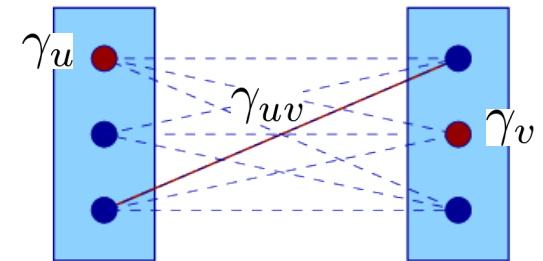
Dual:
$$\max_{\phi} \left[D(\phi) := \sum_{v \in \mathcal{V}} \underbrace{\min_{s \in \mathcal{X}_v} \theta_v^{\phi}(x_v)}_{\gamma_v} + \sum_{uv \in \mathcal{E}} \underbrace{\min_{(s,t) \in \mathcal{X}_{uv}} \theta_{uv}^{\phi}(x_u, x_v)}_{\gamma_{uv}} \right]$$



(1)



(2)

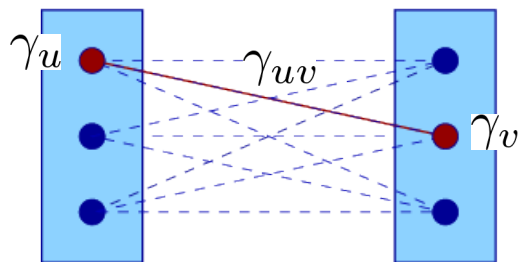


(3)

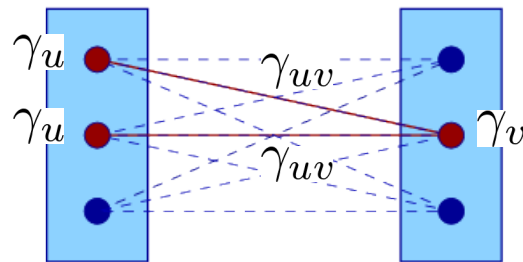
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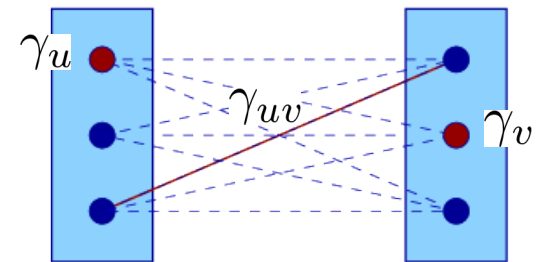
Theorem: If (1) holds for all p/w reparametrized potentials, then the red labels are optimal.



(1)



(2)



(3)

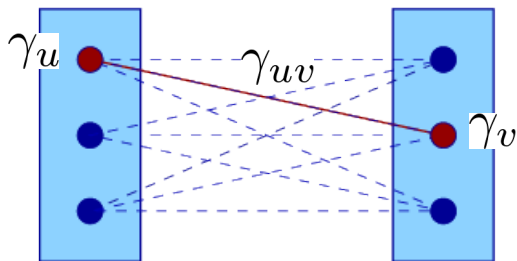
How to Reconstruct a Labeling from a Reparametrization?

Dual:

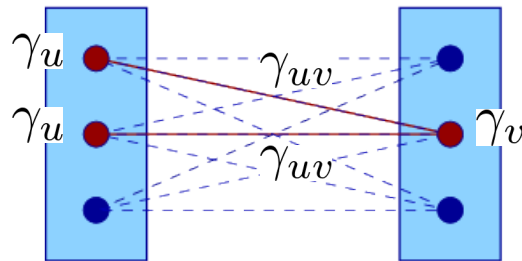
$$\max_{\phi} D(\phi) := \sum_{v \in \mathcal{V}} \underbrace{\min_{s \in \mathcal{X}_v} \theta_v^{\phi}(x_v)}_{\gamma_v} + \sum_{uv \in \mathcal{E}} \underbrace{\min_{(s,t) \in \mathcal{X}_{uv}} \theta_{uv}^{\phi}(x_u, x_v)}_{\gamma_{uv}}$$

Theorem: If (1) holds for all p/w reparametrized potentials, then the red labels are optimal.

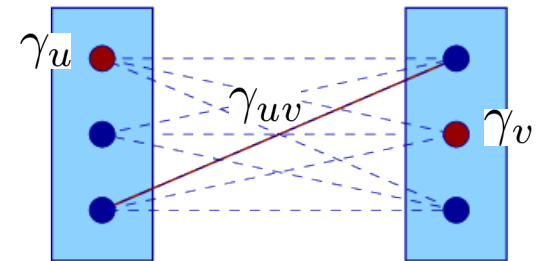
Proof. In general $E(x) \geq D(\phi)$. For red labels $E(x) = D(\phi)$.



(1)



(2)

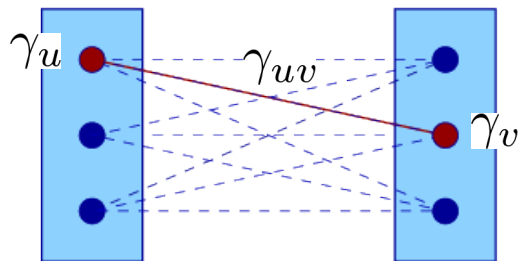


(3)

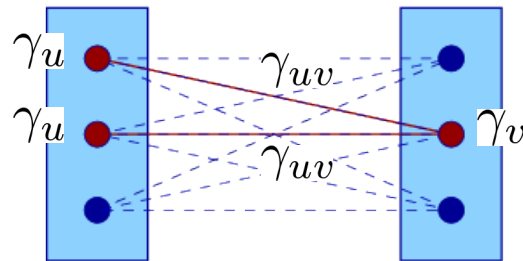
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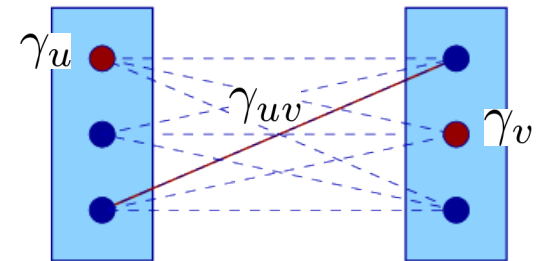
- 1) Can we always find a reparametrization to guarantee (1)?
- 2) How to deal with (2) and (3)?
- 3) Which configurations out of (1-3) are possible in the dual optimum?



(1)



(2)



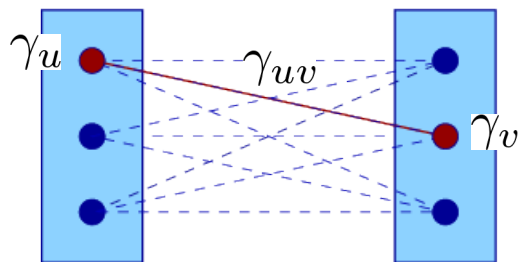
(3)

How to Reconstruct a Labeling from a Reparametrization?

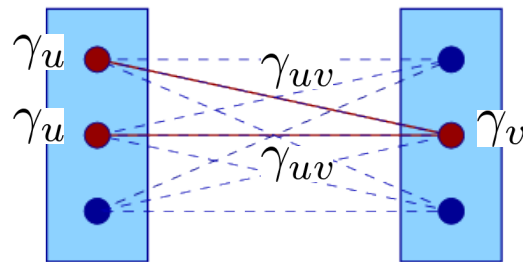
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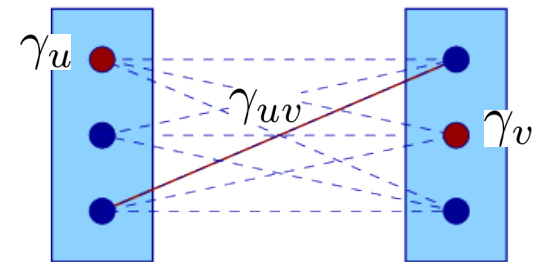
We should go for more theory :)



(1)



(2)



(3)